A Strictly Sofic Mod-Three Gap Suspension and Its Exact Cover Correction

Route-A structural certificate C140

Abstract

We construct an exact nonlattice suspension over the binary mod-three gap shift, a strictly sofic source. Its minimal three-state residue cover has determinant $1-u-v^3$, but the cover has three lifts of the all-zero label fixed point. Correcting that one exceptional orbit at every period gives the intrinsic rational zeta $(1+v+v^2)/(1-u-v^3)$ and an exact primitive label-orbit product. We keep the cover determinant, intrinsic inverse zeta, and any hypothetical natural Fredholm owner distinct. No target or arithmetic factorization is claimed.

1 A strictly sofic source and its cover

Let $X_3 \subset \{0,1\}^{\mathbb{Z}}$ contain the sequences whose finite zero runs between consecutive ones have length in $3\mathbb{Z}_{\geq 0}$; include the all-zero point. It is presented by residue states $0, 1, 2$ and labeled edges

$$0 \xrightarrow{1} 0, \qquad 0 \xrightarrow{0} 1, \qquad 1 \xrightarrow{0} 2, \qquad 2 \xrightarrow{0} 0. \tag{1}$$

The graph is right resolving and strongly connected. The pasts $1, 10, 100$ are synchronizing and end at states $0, 1, 2$. Their residual follower languages are distinct: the shortest words ending at the next 1 from those states are $1, 001, 01$. The follower-set construction of the right Fischer cover must therefore contain these three vertices; (1) realizes exactly them. Thus it is the minimal follower-separated right-resolving cover.

The shift is not of finite type. For any proposed memory L, choose a periodic zero gap $3m+1 > 2L$. The gap violates the mod-three rule, but every length-$(L+1)$ block contains at most one 1. The all-zero block occurs in the all-zero point; a block with one 1 and a, b visible zeros on its sides embeds in an allowed gap of length $3q \geq a+b$. Thus every proposed local block is allowed although the full periodic point is not. No finite local rule can define X_3.

Let u mark label 1 and v label 0. The cover matrix is

$$B(u,v) = \begin{pmatrix} u & v & 0 \\ 0 & 0 & v \\ v & 0 & 0 \end{pmatrix}, \qquad D_{\text{cov}}(u,v) = \det(I-B) = 1-u-v^3. \tag{2}$$

2 The exceptional all-zero orbit

Any label sequence containing a 1 has a unique cover lift: that coordinate forces state 0, and the zero residues propagate in both directions. The all-zero label sequence is different. It is one label fixed point, but has three cover phase lifts forming a single cover orbit of least period three.

Define the intrinsic weighted fixed-point sum explicitly by

$$F_n(u,v) = \sum_{x \in \text{Fix}(\sigma^n|_{X_3})} u^{N_1(x)} v^{N_0(x)}.$$

The ordinary points agree with their unique cover lifts. The all-zero label point contributes v^n for every n, while its three cover lifts contribute $3v^n$ exactly when $3 \mid n$. Therefore, for every $n \geq 1$,

$$F_n(u,v) = \operatorname{Tr} B(u,v)^n + \big(1 - 3\mathbf{1}_{\{3|n\}}\big) v^n. \tag{3}$$

This classification proves (3) without a finite-period inference.

3 Intrinsic zeta and primitive product

Define $\log Z_{140} = \sum_{n\geq 1} F_n/n$ in the total-label-degree completion. The cover part is $-\log(1-u-v^3)$. The exceptional contribution is

$$\sum_{n\geq 1} \frac{v^n}{n} - \sum_{k \geq 1} \frac{3v^{3k}}{3k} = -\log(1-v) + \log(1-v^3) = \log(1+v+v^2).$$

Hence

$$Z_{140}(u,v)=\frac{1+v+v^2}{1-u-v^3},\qquad D_{140}=Z_{140}^{-1}=D_{\mathrm{cov}}\frac{1-v}{1-v^3}. \tag{4}$$

Every periodic label point has a unique primitive label orbit and repetition exponent. Grouping the fixed-point logarithm by that root gives the exact formal product

$$D_{140}(u,v)=\prod_{[\gamma]\ \text{primitive label}}\left(1-u^{N_1(\gamma)}v^{N_0(\gamma)}\right). \tag{5}$$

The regrouping is coefficientwise finite. Equation (4) does not assert that D_{140} is the determinant of a separately constructed natural Fredholm operator on intrinsic label space.

4 Suspension, validation, and boundary

Assign roof times $\tau(1)=1$ and $\tau(0)=\sqrt{2}$, and substitute $u=ze^{-s}$, $v=ze^{-\sqrt{2}s}$. Equation (5) becomes

$$D_{140}(z,s)=\prod_{[\gamma]}\left(1-z^{|\gamma|}e^{-s(N_1+\sqrt{2}N_0)}\right).$$

The label fixed cycles [1] and [0] have lengths 1 and $\sqrt{2}$, proving the roof nonlattice. The positive roof entropy parameter is the unique root of $f(h)=1-e^{-h}-e^{-3\sqrt{2}h}=0$: one has $f(0)=-1$, $\lim_{h\to\infty}f(h)=1$, and $f'(h)=e^{-h}+3\sqrt{2}e^{-3\sqrt{2}h}>0$.

Exact replay through period fifteen contains 969 intrinsic rooted points, 74 primitive label cycles, 60 rooted label-count cells, and 32 primitive cells. The independent checker passes 2,028 assertions, SymPy passes 53 checks, byte replay passes, and all 54 hostile cases (53 repaired-hash and one stale-hash) are rejected. The prefix tests implementation only; (3)–(5) hold at every period by proof.

The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FAIL`), overall `ROUTE_A_EXPLORATORY`. We claim no target divisor, functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, natural self-adjoint or unitary lift, Hilbert–Pólya operator, or Route-B authorization. The literal firewall is

`NO_BAD_EULER_OR_ROOT_NUMBER`.